

Kavirathna Velmurugan

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EDUCATION

SSM Institute of Engineering and Technology | Expected May 2027

B.Tech – Artificial Intelligence & Data Science | Dindigul, Tamil Nadu | CGPA: 7.97

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java

AI/ML & Deep Learning: PyTorch, XGBoost, SHAP, HuggingFace, CLIP, BART-MNLI, QLoRA, LangChain, FAISS, Scikit-learn, Pandas, NumPy

LLM & GenAI: Groq LLaMA, Mistral, Gemini, OpenRouter, RAG Pipeline, Fine-tuning

Web Development: React, Next.js, Node.js, NestJS, FastAPI, Express, Streamlit, Socket.io

Cloud & Databases: AWS (EC2, S3, Lambda), MongoDB, PostgreSQL, Supabase, pgvector

Tools: Git, Docker, Tailwind CSS, Framer Motion, Vite

EXPERIENCE

Skill Hive Innovations

Full Stack MERN Developer Intern | May 2026–Present | Coimbatore

Developing real-world client projects using React, Node.js, MongoDB and Next.js; building Invita — a community management app serving 500+ members with role-based access, WhatsApp/email RSVP, and live event streaming.

Techpuram, Madurai

AI/ML Developer Intern | Nov 2025–May 2026 | Remote

Engineered AST-based code parsing pipeline with FAISS vector embeddings achieving ~40% faster retrieval vs naive text search; built end-to-end ML workflows from raw code ingestion to embedding generation using Python, NLP, and vector databases.

AWS Educate

AI-Powered Cloud Engineer Virtual Intern | Jan 2026–Mar 2026

Earned Grade O (Outstanding) — highest performance grade across 10-week program; built and deployed ML-integrated cloud solutions using AWS EC2, S3, and Lambda.

Gateway Software Solutions

AI & ML Intern | Jul 2025 | Coimbatore

Completed In-Plant Training on AI & ML at ISO 9001:2015 certified company (HP Enterprise delivery partner); gained hands-on experience building ML pipelines in a professional software environment.

PROJECTS

BotStudio — Next.js, Supabase, pgvector, OpenRouter, BetterAuth [bot-studiov1.vercel.app](#)

- Built full-stack RAG chatbot builder — upload PDF/DOCX/TXT → 6-step wizard → AI chatbot embeddable anywhere via script tag.
- Implemented local Xenova embeddings (384-dim), pgvector cosine similarity (top-5 chunks), confidence scoring, and SHAP-style explainability panel.
- Deployed on 100% free tier (Vercel + Supabase) with real authentication, analytics dashboard, and public embed widget.

FarmCredit AI — FastAPI, XGBoost, SHAP, QLoRA, Mistral-7B, Streamlit | [farmcredit-m3o9ke6y7zquszdbitqhe4.streamlit.app](#)

- Built AI crop loan default prediction system achieving AUC-ROC: 0.962, Accuracy: 89.1%, F1: 0.841 on 10,000 synthetic farmer records.
- Fine-tuned Mistral-7B-Instruct using QLoRA/PEFT on ~400 agricultural advisory instruction pairs for farmer-facing advisory generation.
- Implemented SHAP TreeExplainer for per-prediction explainability with risk scoring (Low/Medium/High/Critical) and PDF report generation.

PixelPen AI — React, FastAPI, PyTorch, CLIP, BART-MNLI, Groq | [pixelpen-ai.vercel.app](#)

- Built AI content studio generating 5 coordinated outputs (video script, image prompt, blog, social caption, song suggestions) from a single topic.
- Integrated CLIP for semantic image style retrieval and BART-MNLI for zero-shot tone classification with platform-fit warnings.
- Trained custom PyTorch LSTM on Bach corpus (60 pieces, 20 epochs) for song structure; supports English + Tamil bilingual generation.

City Craft — React, Node.js, Groq LLaMA, Zustand, Framer Motion | [city-craft.vercel.app](#)

- Built browser-based AI startup simulation game set in Tamil Nadu cities with 6 founder archetypes, 3 cities, and 15 civic problems.
- Integrated Groq LLaMA 3.3-70B to generate unique reactions from 20 NPC stakeholders (Mayor, VC, Journalist) based on startup pitch.
- Implemented 5 company actions, real-time expense ledger, reputation system, bankruptcy detection, and localStorage persistence.

DataPilot AI — React, FastAPI, Pandas, Scikit-learn, Groq, MongoDB | [datapilot-ej9z.vercel.app](#)

- Built AI spreadsheet analysis SaaS — upload CSV/Excel (20MB) → auto-detect 7 issue types → AI recommends cleaning → user approves → before/after comparison.
- Implemented Groq LLaMA for NL Q&A; insight card generation, and cleaning explanations with rule-based fallbacks when API unavailable.
- Generated downloadable PDF reports with embedded Matplotlib charts via ReportLab; deployed on Vercel + Render + MongoDB Atlas.

CERTIFICATIONS

- AWS Certified AI Practitioner — Amazon Web Services | Nov 2025 – Nov 2028
- AWS Certified Cloud Practitioner — Amazon Web Services | Mar 2026 – Mar 2029